

Day 09: Skewness and Shape of Data Distribution Understanding Data Behavior

100 Days of Data Science

Why Distribution Shape Matters

Not all datasets are balanced or symmetric. Understanding the shape of data helps us choose correct statistical measures and models.

Distribution shape affects:

- Mean and median values
- Outlier detection
- Machine learning model performance

What is Data Distribution?

Data distribution shows how values are spread across different ranges.

It tells us:

- Where most data lies
- How frequent values occur
- Whether data is balanced or biased

Symmetrical Distribution

In symmetrical distribution:

- Left and right sides are mirror images
- Mean = Median = Mode

Example: Human height in a population often follows symmetric pattern.

Skewed Distribution

Skewness refers to lack of symmetry in data.

There are two main types:

- Positively skewed (Right skewed)
- Negatively skewed (Left skewed)

Positive Skewness (Right Skewed)

Most data is on the left side with a long tail to the right.

Real-life example:

- Income distribution
- House prices

Few very high values pull the tail right.

In right skew:

$$Mean > Median > Mode$$

Negative Skewness (Left Skewed)

Most data is on the right side with a long tail to the left.

Real-life example:

- Marks in an easy exam

Few very low values pull the tail left.

In left skew:

$$Mean < Median < Mode$$

Why Skewness is Important in Data Science

- Affects average calculations
- Influences ML model assumptions
- Determines transformation needs

Handling Skewed Data

Common techniques:

- Log transformation
- Square root transformation
- Outlier treatment

Common Mistakes

- Using mean blindly on skewed data
- Ignoring distribution shape

Final Takeaway

Understanding skewness helps you read data correctly.

Good Data Scientists always check distribution shape before modeling.